Sound, Machine Learning, and Deep Generative Models: A Practical Introduction

The slides are online → https://ktatar.github.io/2026-02_linneaus_audio

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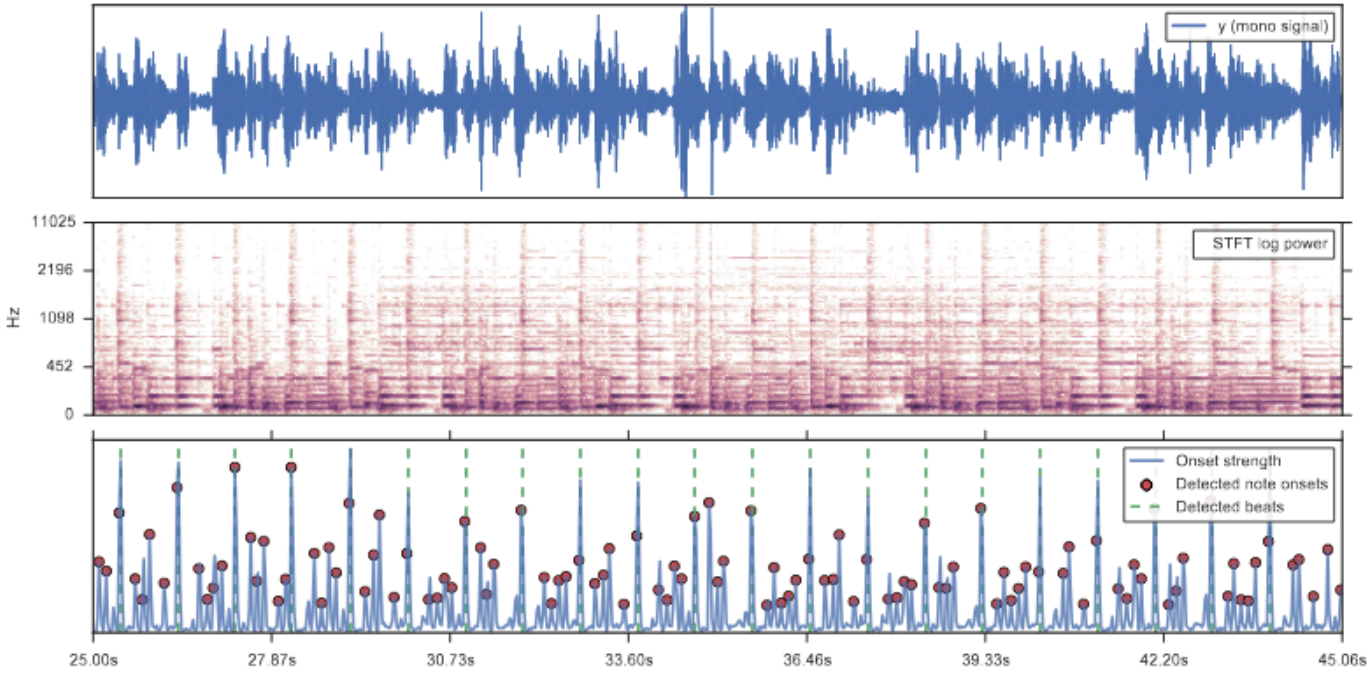
Basics of Sound concepts

- Waveform: dynamics, transients, envelopes
- Frequency: pitch, overtones, timbre components
- Time-frequency transforms: spectrograms, mel spectrograms
- Energy and dynamics over time (envelope, loudness curves etc.)



McFee, Brian, Colin Raffel, Dawen Liang, Daniel PW Ellis, Matt McVicar, Eric Battenberg, and Oriol Nieto. "librosa: Audio and music signal analysis in python." In Proceedings of the 14th python in science conference, pp. 18-25. 2015.

Waveforms and Their Features

-General Introduction:
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Waveform.html
-Phase:
https://images.contentful.com/piwi0eufbb2g/5HdnEvKEB7URTVKDH4iIhf/4ee0fc97b8ac2fe046c195bfc5484228/0_QFVha2lCgyhKjhu0.gif

-Dynamics and Loudness
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Dynamics.html

-Envelope Calculation
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C1/C1S3_Timbre.html

-Onset detection:
https://www.audiolabs-erlangen.de/resources/MIR/FMP/C6/C6S1_OnsetDetection.html

Timbre and its features

FFT based

- Fast Fourier Transform: https://pythonnumericalmethods.studentorg.berkeley.edu/notebooks/chapter24.04-FFT-in-Python.html
- Short-time Fourier Transform: https://www.audiolabs-erlangen.de/resources/MIR/FMP/C2/C2_STFT-Basic.html
- Time vs Frequency Resolution in FFTs: https://support.ircam.fr/docs/AudioSculpt/3.0/co/Window Size.html
- Mel-frequency Cepstral Coefficients
  - Compute pipeline: https://www.researchgate.net/publication/257365356/figure/fig2/AS:297528441491462@1447947716655/Calculation-process-of-MFCC-coefficients.png
  - Example: https://librosa.org/doc/latest/generated/librosa.feature.mfcc.html

Wavelet based Timbre Features

-Constant-Q Transform:
https://librosa.org/doc/latest/generated/librosa.cqt.html

-Variable-Q Transform:
https://librosa.org/doc/latest/generated/librosa.vqt.html

[1] Schorkhuber C, Klapuri A (2010) Constant-Q Transform Toolbox

For Music Processing. In: Proceedings of the 7th Sound and Music Computing Conference (SMC 2010), p. 8. Barcelona, Spain

Music Specific Features

Music specific

- Pitch: https://docs.pytorch.org/audio/stable/tutorials/audio_feature_extractions_tutorial.html#sphx-glr-tutorials-audio-feature-extractions-tutorial-py
- Multipitch (chroma):
  - https://librosa.org/doc/latest/feature.html#spectral-features
- Beat detection:
  - https://librosa.org/doc/latest/generated/librosa.beat.beat_track.html

Machine Learning for Sound and Music Computing

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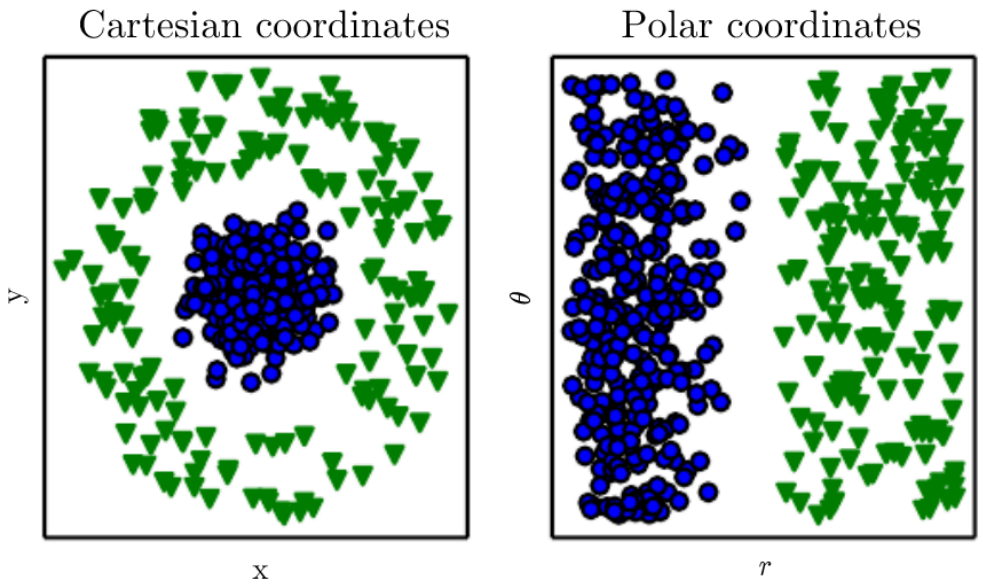
Importance of Feature Representations in Machine Learning



I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.

Dimensionalty reduction

-Principle Component Analysis https://musicinformationretrieval.com/content/10_decomposition/pca.html

-UMAP: Uniform Manifold Approximation and Projection: https://www.kaggle.com/code/adampq/umap-projections-little-shop-of-horrors-audio

-T-distributed stochastic neighbor embedding: https://ml4a.github.io/guides/AudioTSNEViewer/

-Notes on t-SNE: https://dash.gallery/dash-tsne/

Machine Learning

- Supervised Learning
- Unsupervised Learning

Detailed resources: https://www.ibm.com/think/machine-learning#605511093

Supervised Learning - Regression

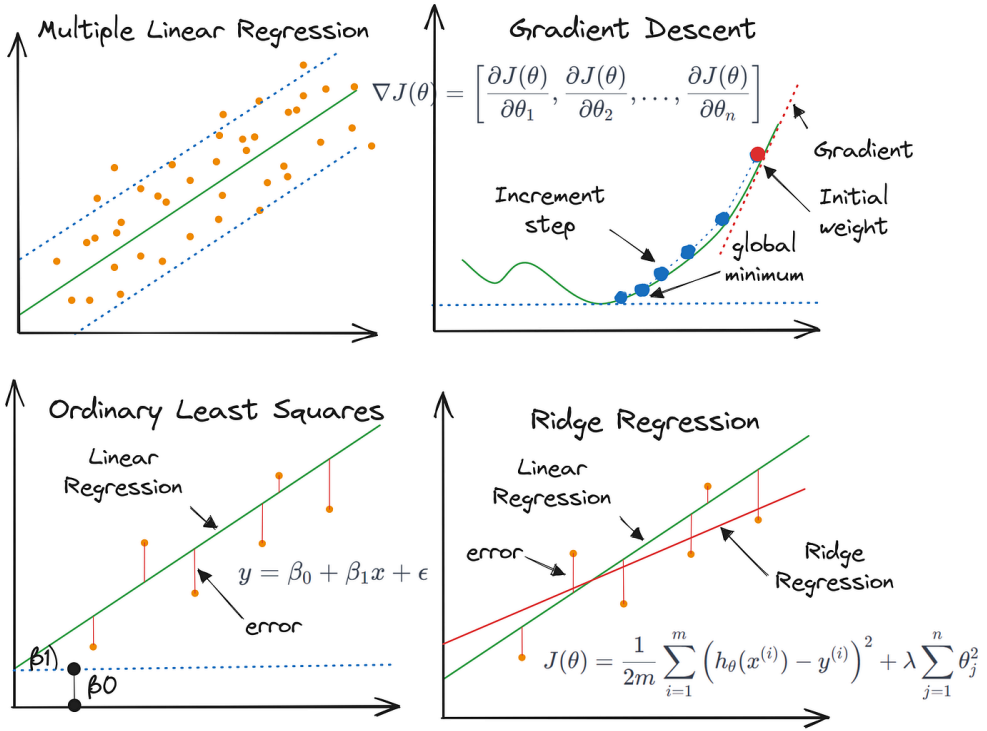


Image source: https://python.plainenglish.io/understanding-multiple-linear-regression-in-machine-learning-58e981ce7747

Supervised Learning - Regression for Interactive Audio

https://www.youtube.com/watch?v=dPV-gCqy9j4

Supervised Learning - Classification

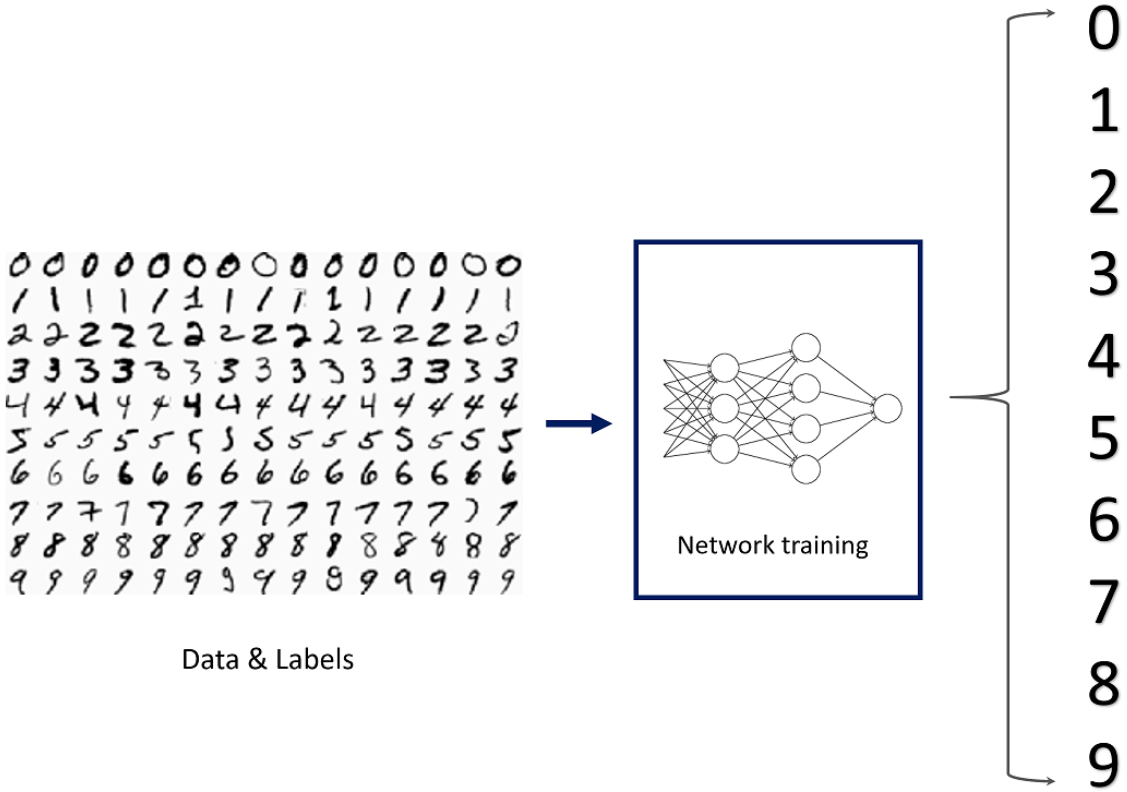


Image source: https://medium.com/data-science/image-classification-in-10-minutes-with-mnist-dataset-54c35b77a38d

Supervised Learning - Audio Classification

Mediapipe: https://ai.google.dev/edge/mediapipe/solutions/audio/audio_classifier

Unsupervised Learning - Clustering

https://colah.github.io/posts/2014-10-Visualizing-MNIST/

Unsupervised Learning - Clustering for Audio

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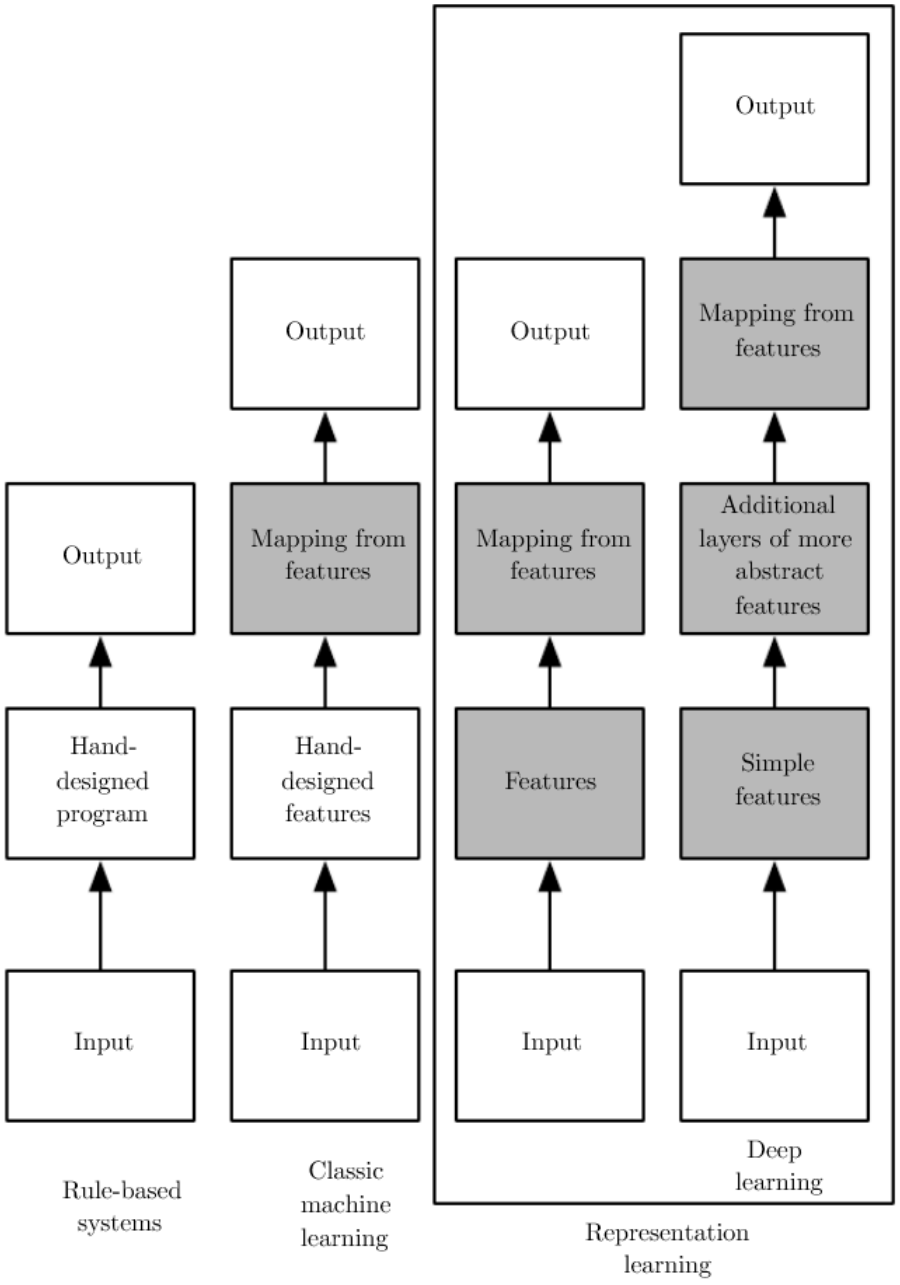
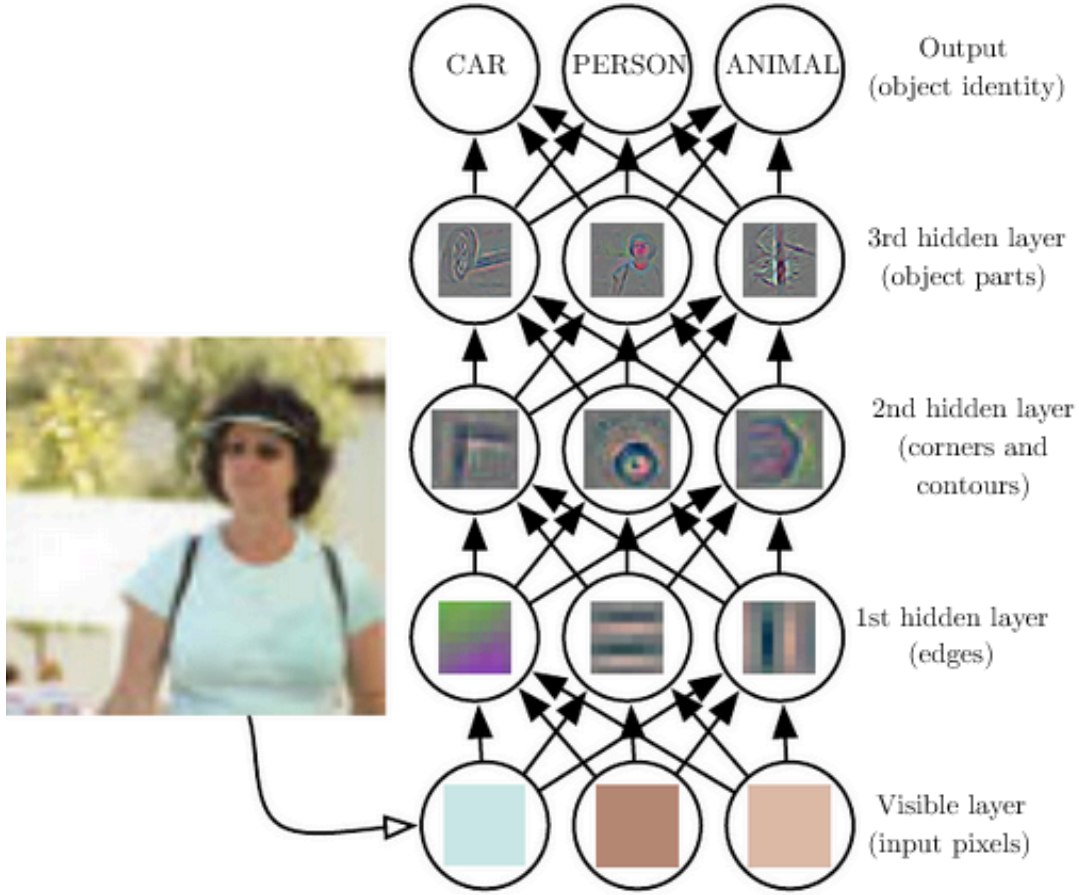
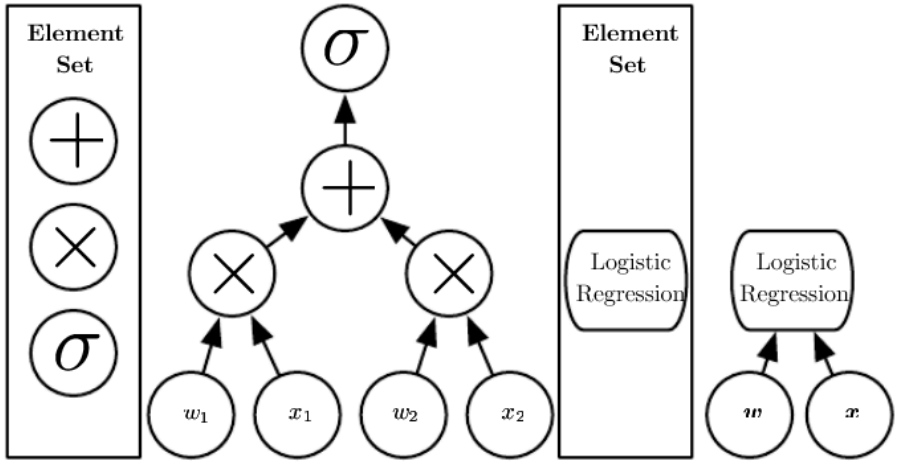
Deep Learning

Two great books to go deeper:

- https://www.deeplearningbook.org/
- https://udlbook.github.io/udlbook/




CAR
PERSON
ANIMAL
Output
(object identity)
3rd hidden layer
(object parts)
2nd hidden layer
(corners and
contours)
1st hidden layer
(edges)
Visible layer
(input pixels)


Element
Set
Element
Set
Logistic
Regression
Logistic
Regression

Deep Learning - Fun Visualizations

- https://adamharley.com/nn_vis/
- https://www.bulovic.at/cnn/

DL based feature representations

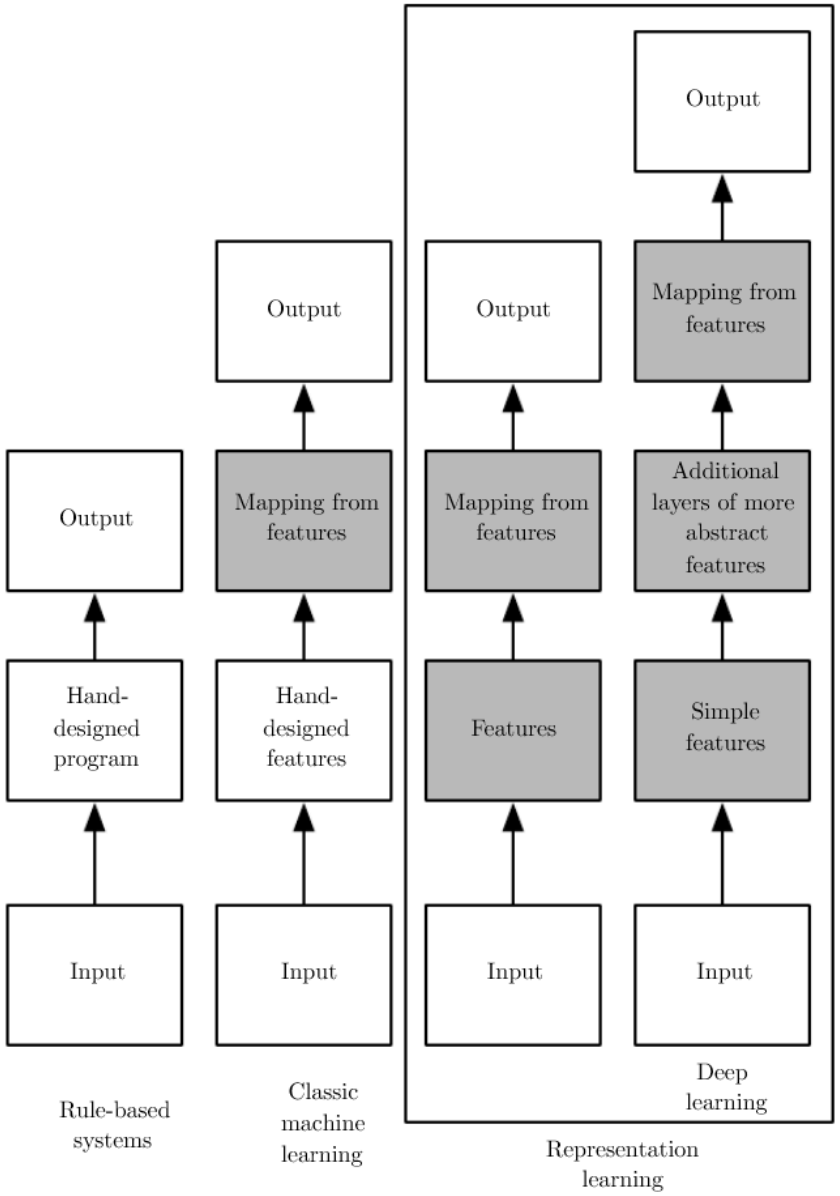
- Deep Embeddings
- Variational Autoencoders
  - https://www.ibm.com/think/topics/variational-autoencoder
  - Live demo: https://www.siarez.com/projects/variational-autoencoder

DL based feature representations - Audio Specific

- Continuous embeddings
  - Stable Audio VAE: https://github.com/Stability-AI/stable-audio-tools
  - SoundStream: https://github.com/haydenshively/SoundStream
- Discrete (for codebooks and tokenization)
  - RQGAN: https://github.com/descriptinc/descript-audio-codec?tab=readme-ov-file

Summary

- Features matter,
- Context specifies the machine learning approach,
- Various ML algorithms can be applied to a single case,
- Simple ML approaches are as applicable as more complex ones.



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Thank you!

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